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High-resolution magnetic stratigraphy across the Oligocene–Miocene boundary in an alluvial–lacustrine succession (Ebro Basin, northeast Spain)

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Abstract

The study of a late Oligocene–early Miocene, 300 m thick section of alluvial and shallow lacustrine sediments in the Ebro Basin, has allowed a high-resolution magnetic polarity stratigraphy for the Oligocene–Miocene boundary to be established. The samples studied include alluvial sandstones and mudstones as well as shallow lacustrine and paludine, micritic and microsparitic limestones. Two-hundred-and-fifty specimens were stepwise thermally or magnetically demagnetized following standard procedures. Magnetic imprints of the distinct lithologies present sharp contrasts, especially for the stability of the secondary components. The VGP data show 18 reversal events, defining a sequence of 16 polarity intervals. Lateral and vertical continuity of the sequences studied has enabled a correspondence between the local paleomastological biozones and the magnetostratigraphic succession to be established. The high-resolution chronology of this magnetostratigraphic succession provides valuable information that allows the non-marine European biochronological scales to be correlated to the global polarity time scale (GPTS).

The magnetic polarity zonation of the section has been correlated to the GPTS using the distinct sequence and pattern polarity reversals as well as by considering the fossil mammal assemblages. As a preliminary result, chrons 8, 7A, 7 and 6C have been identified.

1. Introduction

Despite the fact that Paleogene and Neogene paleomastological reference levels, 'zones' and 'stages' have been established long ago (Thaler, 1972; Mein, 1975) these biochronological units are often not well correlated with the standard stratigraphic scales (see for instance Schmidt-Kittler, 1987; Steininger et al., 1989; Prothero and Berggren, 1992). Scarcity of well-developed lateral relationships between marine and continental sequences in most of these basins often precludes a direct, first-order correlation. Moreover, geochronology is not always possible. As a conse-

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quence, magnetostratigraphic studies of non-marine, fossil mammal-bearing successions seem to be the best approach to establish an accurate correlation between marine and non-marine biochronological scales. Recent contributions point in that direction (Burbank et al., 1992; Prothero and Berggren, 1992; Leveque, 1992, 1993a, b; Parés et al., 1993; Leveque and Sen, 1994). The interest in this approach results from the fact that a complete understanding of the global changes which affected the Earth will only be possible when a global geochronology has been achieved.

It is emphasized that one of the major con-

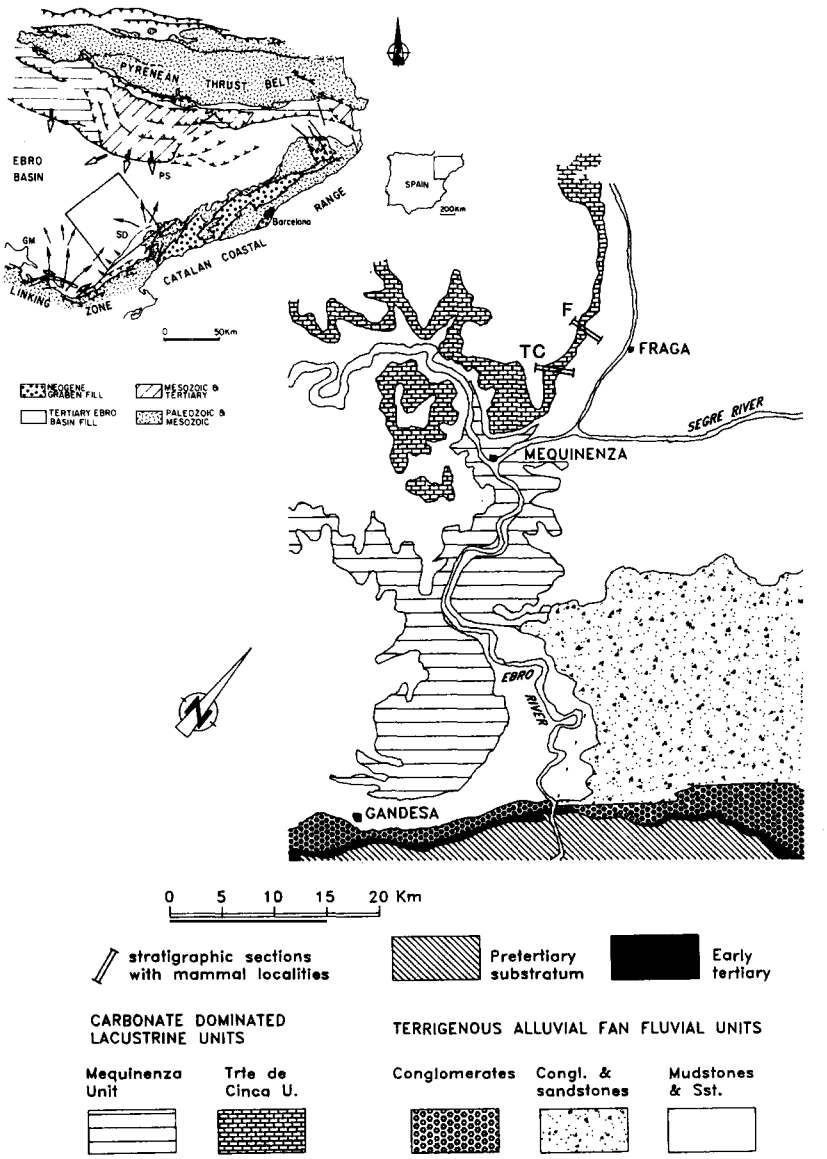


Fig. 1. Geological setting of the Ebro Basin in northeast Spain and simplified facies map of the southeastern Ebro Basin, showing distribution of the main depositional units. The terrigenous alluvial fan to fluvial units correspond to three major systems: the Scala Dei (SD), Guadalope–Matarranya (GM) and Pyrenean (P) systems; the lower lacustrine units belong to the Los Monegros System.

straints to be considered in producing an accurate biochronological–magnetostratigraphic correlation is its application, wherever possible, to continuous, thick fossil mammal-bearing sequences suitable for obtaining long, testable magnetostratigraphic records of the sequences studied. The fulfilment of these requirements will make the comparison of the obtained magnetostratigraphic scales with the global polarity time scale (GPTS) easier and more reliable, and it is particularly important for the Oligocene–Miocene boundary owing to the poor definition that exists for:

(1) the Oligocene–Miocene boundary (Bizon and Muller, 1977; Bolli et al., 1985; Martini, 1988; Steininger et al., 1989; Harland et al., 1990; and see Berger (1992) for a detailed review of European case studies);
(2) the precise dating of the successions where marine and non-marine late Oligocene–early Aquitanian coexist (Anglada, 1972; Catzigras and Glintzbeockel, 1972; Hugueney and Truc, 1976a and b; Alvinerie et al., 1977; Aguilar, 1982; Hugueney and Ringeade, 1989).

Previous studies carried out in the Cenozoic sequences of the southeastern Ebro Basin (Cabrera, 1983) established that the above-mentioned constraints were accomplished and that diverse late Oligocene–early Miocene mammal fossil assemblages follow one another through a continuous stratigraphic sequence up to 500 m thick. This stratigraphic continuity, and the relatively large number of fossil mammal localities allowed for a preliminary micromammal biozonation (Agustí et al., 1988). This biozonation was based not only on the evolutionary stages of the recorded taxa (mostly Eomyidae, Gliridae, Theridomyidae and Cricetidae), but also on their stratigraphic distribution in continuous and correlated sequences (Cabrera, 1983; Agustí et al., 1988). Thus, the southeastern Ebro Basin is one of the unusual cases in Europe where a continuous non-marine late Oligocene–early Miocene stratigraphic sequence can be zoned by means of mammals with such potential detail (Agustí et al., unpublished work).

Hence, the southeastern Ebro Basin has an extraordinary potential for being able to locate the GPTS Oligocene–Miocene boundary in relation to local mammal biozones. This paper is involved in establishing this boundary and in locating the previously determined paleomastological biozones in the GPTS. Moreover it is a contribution to a research program that focuses on the correspondence of the non-marine biozonations established in the Iberian Cenozoic basins with the GPTS.

2. Geological setting: the Ebro Basin

The Cenozoic Ebro Basin developed in the northeastern Iberian plate as a result of the tectonic evolution of the Southern Pyrenean Foreland (Puigdefàbregas et al., 1992; Vergés et al., 1992). It is bounded to the north by the Pyrenees, to the southwest by the Iberian Range and the Linking Zone and to the southeast by the Catalan Coastal Ranges (Fig. 1). The basin is asymmetrical with the main sedimentary depocenters located in the northern basin zones, where the pre-Tertiary substratum ranges from depths of 3000 to 5000 m (Riba et al., 1987).

From the Paleocene until the late Eocene, the Ebro Basin was connected with the Atlantic Ocean and marine environments were widespread in the northern areas of the basin (Souquet and Puigdefàbregas, 1986; Riba et al., 1987). From the latest Eocene–early Oligocene to the late Miocene, the basin became closed and was characterized by non-marine sedimentation (Cabrera et al., 1985; Cabrera and Sáez, 1987; Riba et al., 1987; Pérez, 1989; Anadón et al., 1989a, b; Arenas, 1993).

The main influence on the evolution of late Oligocene–early Miocene sedimentary infilling of the Ebro Basin resulted from subsidence triggered by the Pyrenean orogen and the major paleogeographic changes caused by the southward emplacement of successive Pyrenean thrust sheets. Nevertheless, the tectonic activity in the Iberian and Catalan Coastal Ranges also noticeably influenced the general facies arrangement (Anadón et al., 1985, 1989a, b).

3. Late Oligocene–early Miocene depositional framework of the southeastern Ebro Basin

As a consequence of tectonic activity, well-developed alluvial fan systems formed along the margins of the eastern Ebro Basin. Thus, the late Oligocene–early Miocene main depositional framework in the study zone consisted of distributive alluvial systems fed from the tectonically active surrounding ranges and laterally related to inner-marginal, carbonate-dominated to evaporitic, shallow lacustrine zones. Alluvial and lacustrine systems were closely related and correlative advances and retreats of both kinds of systems took place during their evolution (Figs. 1 and 2).

Four major sedimentary systems are considered in the late Oligocene–early Miocene depositional framework of the study zone: (1) Scala–Dei alluvial system; (2) Guadalope–Matarranya alluvial system; (3) Pyrenean alluvial system; (4) Los

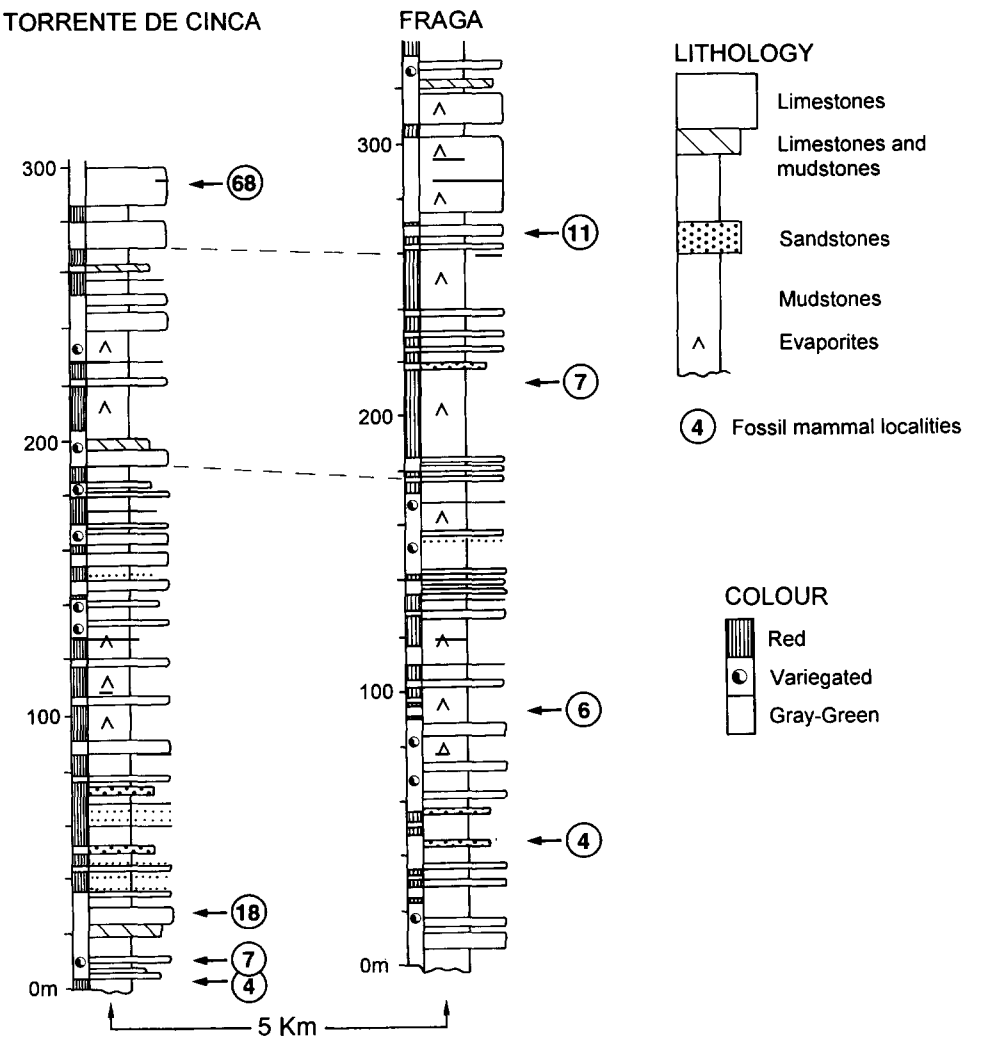


Fig. 2. Stratigraphic logs of the Torrente de Cinca and Fraga sections showing their correlation and the distribution of the fossil mammal localities.

Monegros lacustrine system (Fig. 1) (Colombo, 1980, 1986; Cabrera, 1983; Cabrera et al., 1985; Anadón et al., 1989a, b).

The sequences of the Los Monegros lacustrine System developed mainly in inner basinal zones and display vertical and lateral relationships with all of the above-mentioned alluvial–fluvial systems (Cabrera, 1983; Cabrera et al., 1985; Anadón et al., 1989a, b). In the area studied, thick lacustrine sequences span from the Oligocene to early Miocene. Further west the lacustrine deposits had a longer time span (Oligocene–middle Miocene) and exhibit a more evaporitic character. The lacustrine facies assemblages and their stratigraphic relationship with the alluvial facies show that the Los Monegros system record the development of a shallow swamp–marsh and lake complex. The lacustrine system experienced successive advances and retreats, recorded in a 500 m section of lacustrine carbonates alternating with mudstones and sandstones in the inner basinal zones (Cabrera et al., 1985; Cabrera and Sáez, 1987; Anadón et al., 1989b).

4. The Torrente de Cinca section

Two main well-exposed sequences with several mammal sites are located in the studied area: the Torrente de Cinca section and the Fraga section. The distance between these two sections is 5 km (Fig. 2).

The bottom of the Torrente de Cinca section reported here is located at km 114 of the Barcelona to Zaragoza highway while its top is at km 109. The 300 m thick section consists of alternating alluvial terrigenous facies (red mudstones and litharenitic sandstones) and lacustrine–palustrine carbonate facies (microsparitic limestone, marly limestones, strongly root-bioturbated limestones, chert and anhydrite and gypsum nodule-bearing limestones). The percentage of terrigenous facies is nearly 60% of the section thickness, while that of the carbonate facies is approximately 40%. Sandstones occur only in the lower half of the section where they make up tabular beds or channel shaped bodies up to 1.5 m thick (Fig. 2).

Single beds and packets of lacustrine carbonates, ranging from a few centimeters to several meters in thickness are distributed along the section, except in its mudstone-dominated upper middle part. The thickest carbonate beds occur at the top of section, where two bodies up to 18 m thick have been recorded. Evaporite facies, mainly formed by nodular secondary gypsum, occur mainly in the middle part of the section, while chert nodule-bearing facies is restricted to a 10 m thick lacustrine carbonate assemblage located at 200 m from the base of the section.

Lacustrine influence on sedimentation of the Torrente de Cinca section was nearly continuous, except during the development of the red mudstone-dominated facies assemblage (from 220 to 240 m), which was deposited mainly under subaerial, dry mud flat paleoenvironmental conditions. The thickness of this terrigenous-dominated sedimentary episode is here relatively low when compared with that recorded in the neighboring sequences located more to the north (i.e. Fraga section) and to the south, closer to the influence of the Pyrenean and the Guadalope–Matarranya and Scala–Dei systems, respectively (Fig. 1). The alluvial spreading recorded by this facies assemblage is of major significance (Cabrera, 1983; Cabrera et al., 1985; González, 1989; Anadón et al., 1989b). A later lacustrine spreading episode developed in the basin, resulting in the deposition of the upper thick carbonate packets. This environmental change also had a wide regional significance.

The importance of the Torrente de Cinca section arises from the fact that it contains several fossil mammal localities and can be correlated with other mammal sites in the Fraga W section (Figs. 1 and 2). This correlation is based on lithostratigraphic and photogeological mapping of some noticeably continuous carbonate lacustrine units, which can be traced from one section to the other (Cabrera, 1983; Agustí et al., 1988). The established mammal biozones in the section (*Eomys* aff. *major*, *Rhodanomys transiens* and *Rhodanomys schlosseri* biozones) enabled a late Oligocene–early Miocene age to be attributed to the section and to propound a biostratigraphic zonation.

5. Field and laboratory procedures

Along the Torrente de Cinca section, field sampling was accomplished with electrical and petrol-powered portable drilling machines equipped with a non-magnetic drill bit with diamond crown. Cylindrical cores of 2.54 cm diameter by, approximately 10 cm length were obtained. Sometimes in poorly consolidated layers (mudstones) oriented blocks were collected and drilled in the laboratory.

Generally, three samples per layer at 3 m intervals were taken. A shorter interval was used at specific levels (carbonate lithofacies). About 280 samples were obtained from a total of 108 sites. Thirty samples were rejected during preliminary laboratory preparation or refused after analyzing the results of thermal and AF demagnetization because of low intensity of NRM.

The samples were treated at the Paleomagnetic Laboratory of the Institute of Earth Sciences (CSIC, Barcelona), and measurements included the following.

(1) Natural remanent magnetization (NRM) and remanent magnetization during magnetic cleaning. These measurements were carried out with a three-axis GM400 cryogenic magnetometer (CCL).

(2) Magnetic susceptibility, obtained with the susceptibilitymeter Kappabridge KLY-2 (Geofizika Brno), operating at 300 A m^{-1}.

(3) Thermal demagnetization (demagnetizer TSD-1 Schonstedt) using 50°C incremental steps up to 680°C.

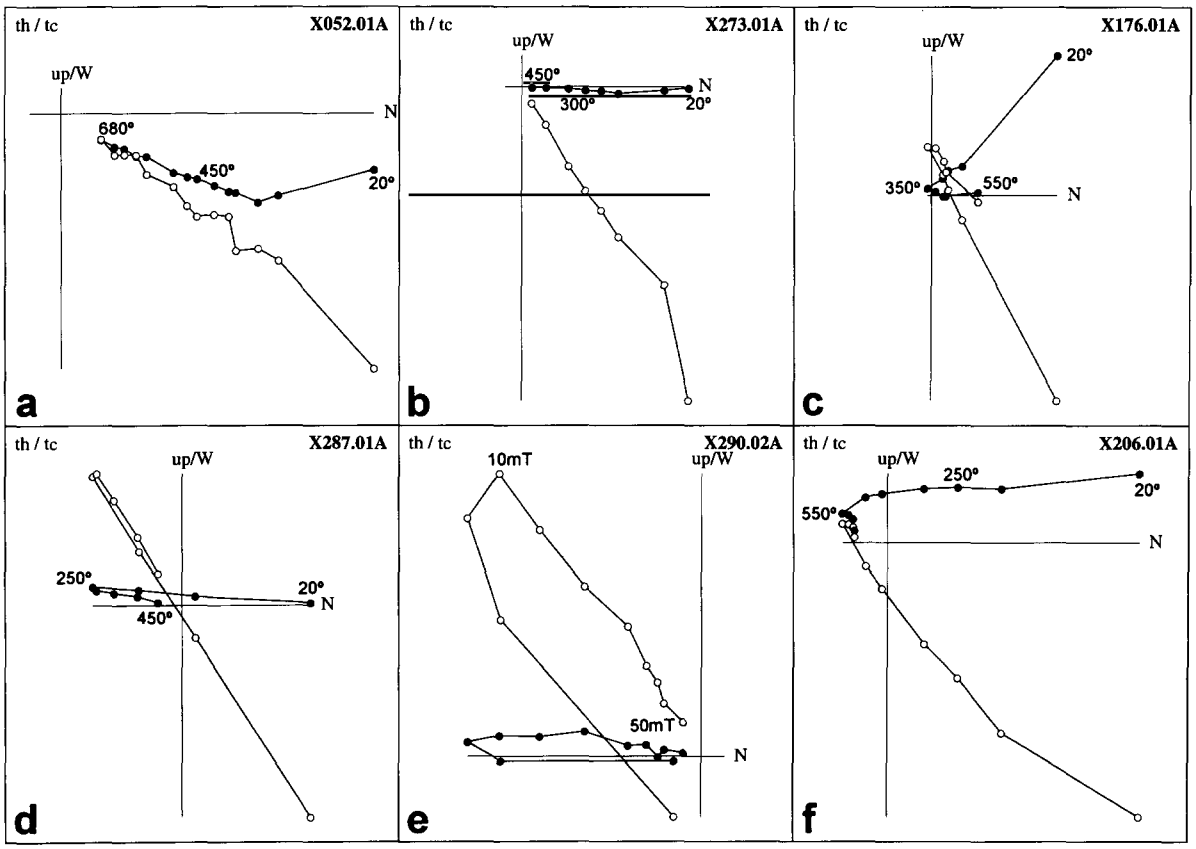


Fig. 3. Orthogonal Zijderveld plots for six representative samples from the Torrente de Cinca section. Full (open) symbols are vectors on the horizontal (vertical) plane: (a)–(c) detritic lithofacies; (d)–(f) carbonate lithofacies.

(4) Alternating field demagnetization (demagnetizer GSD-5 Schonstedt) with magnetic fields of 0, 5, 7.5, 10, 15, 20, 25, 30, 35, 40 and 50 mT.

Thermal demagnetization was found to be more effective above all carbonate lithofacies as will be shown below.

Data obtained in the laboratory have been represented in Zijderveld (1967) diagrams, allowing primary and secondary ferromagnetic components to be differentiated and their demagnetization intervals to be established (Fig. 3).

Taking into account the results of the demagnetization 101 sites of the 108 initially sampled gave reliable results and were used for the VGP Paleolatitude calculation.

6. Initial susceptibility and natural remanent magnetization

From the 250 suitable samples, 197 representative ones were analyzed and three groups with different magnetic behavior have been recognized.

(1) Low bulk susceptibility (-20 to 40×10^{-6} SI) and NRM intensity (7×10^{-6} to 10^{-4} A m^{-1}). This group comprises limestones with anhydrite nodules, bioturbated limestones and micritic limestones, almost all of the marly limestones and a few sandstones.

(2) Intermediate susceptibility ($40–180 \times 10^{-6}$ SI) and NRM intensity (10^{-4} to 6×10^{-4} A m^{-1}). Sandstones with anhydrite nodules, a few marly limestones and a portion of sandstones and red mudstones are included within this group. The maximum susceptibility values ($100–130 \times 10^{-6}$ SI) are mainly found in the sandstones.

(3) High susceptibility ($190–250 \times 10^{-6}$ SI) and NRM intensity (6×10^{-4} to 4×10^{-3} A m^{-1}), particularly the sandstones and red mudstones.

7. Natural remanent magnetization directions

NRM vectors cluster around the present-day field direction before any treatment (Fig. 4). The mean direction is 351/50, showing an important overprint of the recent geomagnetic field. As will be shown, both thermal and AF demagnetization procedures successfully removed the present-day geomagnetic field component, although the former method proved to be more effective.

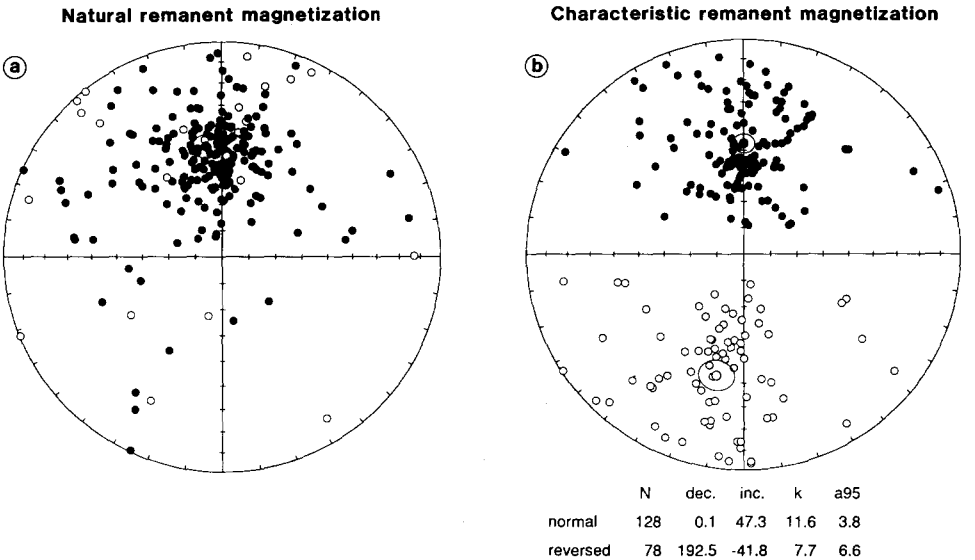


Fig. 4. Equal area lower hemisphere stereographic projection of (a) NRM directions and (b) ChRM directions for the Torrente de Cinca section.

Because of the paleomagnetic differences of the collected lithologies, they will be treated and described separately.

7.1. Carbonate lithofacies

Almost all of the lacustrine–palustrine carbonates display relatively stable behavior and show a progressive decrease in magnetization during magnetic cleaning (Fig. 3). For these lithofacies, both thermal and AF demagnetization have been used.

Gypsum and anhydrite nodules in carbonate lithofacies induce a considerable instability in samples preventing the determination of the primary magnetization. We attribute such instability to the chemical changes of the anhydrite grains induced by beating during thermal demagnetization. This fact has been observed in a temperature interval between 250°C and 400°C.

Chert nodule-bearing limestones are characterized by high NRM values and they show excellent magnetization stability and decay.

Strongly root-bioturbated limestones, micritic limestones and marly limestones show two superimposed magnetic components, a feature which has also been observed in the non-carbonate lithofacies. A low-temperature component is removed at 250°C and corresponds to the present-day geomagnetic field, while a high-temperature component is defined at 300°C. These lithologies present a stable magnetization till 450°C, sufficient to determine the characteristic remanent magnetization. The demagnetization temperature

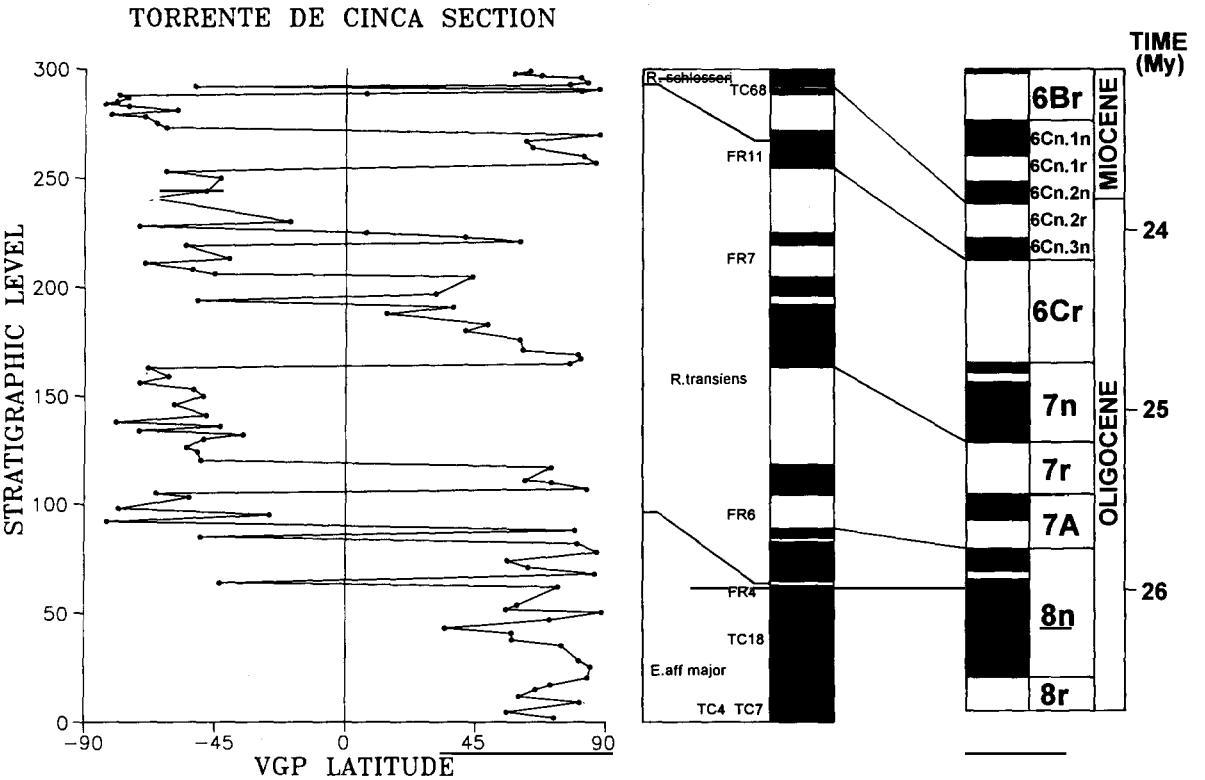


Fig. 5. VGP latitude of the Torrente de Cinca section and correlation to the GPTS (Cande and Kent, 1992). TC (Torrente de Cinca section) and FR (Fraga section) indicate the positions of the mammal sites of Fig. 2.

suggests a mineral with T_{ub} around 450°C and in some cases even less. Such a mineral may be magnetite with a small amount of titanium, but some minor of pyrrhotite may be present to account for the T_{ub} about 300°C.

7.2. Detritic lithofacies

Terrigenous lithofacies like sandstones and red mudstones (possibly with anhydrite) have a stable magnetization, requiring temperatures higher than 400°C to determine the characteristic remanent magnetization (Fig. 3).

Most of the samples show two components:

(1) a low-temperature secondary component with the same characteristics that those recorded in the carbonate lithofacies;
(2) a more stable component with a blocking temperature above 600°C, corresponding to the hematite spectrum.

Magnetic cleaning of the detrital lithofacies has been carried out using only thermal treatment, since AF treatment is unable to remove the hematite magnetization.

8. Discussion: magnetostratigraphic sequence and chronology of the sections studied

The characteristic remanent magnetization (ChRM) was computed for each specimen using Principal Component Analysis (Kirschvink, 1980). The stereographic projection of both NRM and ChRM directions reveals that demagnetization procedures have successfully removed the secondary component of the samples (Fig. 4).

Normal and reversed directions strike along a north–south axis, revealing no significant rotation relative to north. Mean directions are: dec. = 0.1°, inc. = 47.3° and dec. = 192.5°, inc. = −41.8°.

This result coincides with previous studies carried out in late Oligocene and lower–middle Miocene sediments from the northern and southeastern Ebro Basin and the Calatayud–Daroca Basin (Dijksman, 1977; Turner et al., 1984; Friend et al., 1988–1989).

For each sampling level, the mean virtual geomagnetic pole (VGP) latitude has been calculated, enabling the establishment of a scale of normal and reversed magnetozones. Thus the polarity pattern for the section is derived from the position of the pole paleolatitude for each site.

Paleolatitude of the VGP calculated for the 101 suitable sites has been plotted in Fig. 5 for each sampling level. Our data show 18 reversals that define 16 polarity intervals. These also include five short subchrons based on one sample level. All of these short subchrons have been considered also on the basis of: (1) the high quality of the paleomagnetic signature and (2) each sampling level comprises two or three consistent samples from different cores.

The magnetic polarity zonation of the entire section was correlated to the GPTS using the distinct sequence and pattern of polarity reversals after considering the approximate age estimation of the Torrente de Cinca and Fraga sections based on the fossil assemblages (Agustí et al., 1988). Three local mammal biozones were preliminary established in the section (*Eomys* aff. *major*, *Rhodanomys transiens* and *Rhodanomys schlosseri*) that cover the latest Oligocene and earliest Miocene (late Chattian–early Aquitanian) time span. The lower mammal sites (TC4, TC7 and TC18), which have been ascribed to the *Eomys* aff. *major* zone (Agustí et al., 1988), correspond to the lower normal polarity zone. The Fraga 4 (FR4) mammal site, located in the nearby Fraga section, may also be correlated to this polarity zone. The FR6, FR7 and FR11 mammal sites are also located in the Fraga section. These sites, which are ascribed to the *Rhodanomys transiens* local zone (Agustí et al., 1988) are placed stratigraphically between the fossil localities on top and at the bottom of the Torrente de Cinca section and are coeval with several polarity changes. The top of the Torrente de Cinca section, located in a normal event, contains the TC68 mammal site, which corresponds to the *Rhodanomys schlosseri* local zone (Figs. 2 and 5).

The exact correspondence of some of the local southeastern Ebro Basin biozones to the general late Oligocene paleomastological reference levels is difficult to achieve, owing to the diverse methodological approaches used to define both kinds of paleomastological subdivisions (Agustí et

al., 1987, 1988; Agustí and Moya, 1991; Burbank et al., 1992). The definition of the successive general reference levels on the basis of the evolutionary stage reached by diverse taxa introduces some elements of uncertainty, especially when considering the use of fossil mammal taxocenosis that come from disperse, non-correlatable localities in distinct separated basins and karst infilling fissures.

All the mammal sites which correspond to the *Eomys* aff. *major* local zone (Chattian) have been found in a long normal polarity zone of the lower part of the Torrente de Cinca section. This distinctive long period of normal polarity at the bottom of the section has been used as a major anchor point for the GPTS fit and is tentatively correlated to chron 8n of the late Oligocene up section. From this point the pattern of the local Torrente de Cinca section magnetostratigraphic scale contrasted with the Cande and Kent (1992) GPTS proposal, suggests the occurrence of chrons 7A, 7r, 7n, 6Cr and 6Cn (Fig. 5). This correlation suggests that the long magnetic reverse event in the upper part of the Torrente de Cinca section corresponds to chron 6Cr, an especially long and characteristic late Oligocene reverse event. As a consequence, the normal event on top of the Torrente de Cinca section corresponds the 6Cn chron, presumably the 6Cn.2n subchron. According to these data, and following the proposal of Cande and Kent (1992), the Oligocene–Miocene boundary would be located at the top of 6Cn.2r, which in our case means that this boundary would be located at the base of the normal event on top of the Torrente de Cinca section. This normal event includes the TC68 mammal site (*R. schlosseri* local zone).

We observe that not only all of the GPTS chrons have been recorded along the studied section, but also some subchrons. Some of these subchrons can be tentatively correlated with those established by Cande and Kent (1992) (i.e. the 6Cr normal subchron and the 8n reversed subchron near FR4). Others are not reflected in the oceanic record of this reference scale (i.e. the 6Cn.2n reversed subchron and the 8n reversed subchron near the top of Chron 8n) (Cande and Kent, 1992). Further investigation is required to confirm these events although our paleomagnetic record is high in quality and the ChRM directions have been unambiguously determined. Despite their significance (artifacts, selective remagnetizations, true reversal), the relatively rapid sedimentation rate (11 cm kyear^{-1}) in the Torrente de Cinca area may account for the recording of short subchrons otherwise unrecognized in the oceanic record.

The correlation here proposed between the GPTS and the local Torrente de Cinca magnetostratigraphy has been combined with the preliminary paleomastological data (Agustí et al., 1988). A fossil mammal locality (FR11) located in the normal event here attributed to 6Cn.3n has delivered *R. transiens* (Agustí et al., 1988). In addition, another fossil mammal locality with *R. schlosseri* (TC68) occurs in the normal event attributed to the 6Cn.2n subchron.

Following our magnetostratigraphic correlation, FR11 and TC68 are, respectively, Oligocene and early Miocene in age. *R. transiens* from FR11 is a rodent species which characterizes several late Oligocene and possibly earliest Miocene assemblages of the paleomastological reference level MP30 (Schmidt-Kittler, 1987) and the lower MN-1 zone (Hugueney and Ringeade, 1989). *R. schlosseri*, recorded in TC68, characterizes the upper MN1 zone which has often been attributed to the early Miocene (Mein, 1975; Hugueney and Truc, 1976a, b; Aguilar, 1982; Agustí et al., 1987, 1988; Hugueney and Ringeade, 1989; Agustí and Moyà, 1991). Thus the ages deduced from the paleomastological assemblages preliminary record from the TC68 and FR11 localities do not contradict our correlation between the local magnetostratigraphic section and the GPTS (after Cande and Kent, 1992).

9. Conclusions

A correlation of the magnetostratigraphic zonation of the Torrente de Cinca section with the GPTS of Cande and Kent (1992) is proposed on the basis of the studies carried out on thick, continuous and correlatable late Oligocene–early

Miocene sequences in the southeastern Ebro Basin. The occurrence of chrons 8, 7A, 7 and 6C is suggested.

The preliminary paleomastological biozonation previously established for these sequences (Agustí et al., 1988) has been correlated with the magnetostratigraphic scale, and the results are not contradictory with the proposed correlation, with the occurrence of rodent taxa characteristic of the latest Oligocene–early Aquitanian at the top of the sequences studied.

One of the major interests of this contribution results from the potential possibility to define biochronological subdivisions based on paleomastological assemblages with well-known stratigraphic relationships within long local magnetostratigraphic sequences, suitable to be correlated confidently with the GPTS.

A certain degree of uncertainty remains in the very precise relationships between the proposed mammal biozone and the magnetozone boundaries owing to the existence of barren fossil mammal intervals (Figs. 2 and 5). New fossil mammal localities must be found in order to overcome this lack of information. Moreover, a further extension of the paleomagnetic studies to the stratigraphic sequences which overlie and underlie the Torrente de Cinca section will contribute to the verification of the GTPS correlation proposed here and to clarification of the uncertainties that still exist for the late Oligocene–Miocene boundary determined on the basis of the European paleomastological fauna.

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